

Demonstrating InflatableMod: Untethered and Reconfigurable Inflatable Modules for Tabletop-sized Pneumatic Physical Interfaces

Takafumi Morita
The University of Tokyo
Tokyo, Japan
morita@xlab.iii.u-tokyo.ac.jp

Ziyuan Jiang
The University of Tokyo
Tokyo, Japan
jiang@xlab.iii.u-tokyo.ac.jp

Kanon Aoyama
The University of Tokyo
Tokyo, Japan
aoyama@xlab.iii.u-tokyo.ac.jp

Ayato Minaminosono
Shibaura Institute of Technology
Tokyo, Japan
nb20108@shibaura-it.ac.jp

Yu Kuwajima
Shibaura Institute of Technology
Tokyo, Japan
nb21105@shibaura-it.ac.jp

Naoki Hosoya
Shibaura Institute of Technology
Tokyo, Japan
hosoya@sic.shibaura-it.ac.jp

Shingo Maeda
Tokyo Institute of Technology
Tokyo, Japan
maeda.s.ao@m.titech.ac.jp

Yasuaki Kakehi
The University of Tokyo
Tokyo, Japan
kakehi@iii.u-tokyo.ac.jp



Figure 1: InflatableMod. (A) InflatableMod consists of Phase-change Control Module and Pouch Module, (B) Linear liquid-to-gas phase change actuator control, (C) Reconfigurable inflatable lump, (D) Continuous data physicalization on a map, (E) Silent biomimetic soft actuator.

ABSTRACT

Inflatable systems have been attracting attention in the field of interaction design. Conventional tabletop-sized pneumatic systems tend to be complex because they require bulky and noisy equipment. Therefore, several liquid-to-gas phase change actuators that use vaporization have been proposed. But these actuators have problems with controllability, reusability, and reconfigurability. In this study, we propose InflatableMod, novel inflatable modules based on the efficient control of liquid-to-gas phase change actuators. These are designed with a compact circuit that has a liquid transfer function to feed the required amount of low-boiling-point liquid into the pouch and a heating function to inflate the pouch by the volume change. This approach allows for a compact, silent, and untethered inflatable system. It is also possible to create an untethered and reconfigurable multi-inflatable system because each module is synchronized. In this paper, we demonstrate the basic operation and some applications with InflatableMod.

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CCS CONCEPTS

• Human-centered computing → Interactive systems and tools.

KEYWORDS

Programmable Materials, Inflatables, Multi-inflatables, Pneumatic, Liquid-to-gas Phase Change Actuators, Electrohydrodynamics.

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1 INTRODUCTION

Shape-changing interfaces have long attracted attention because they can dynamically change the shape of substantial objects. Particularly, inflatable shape-changing interfaces have the advantage of being freely changeable in volume. Various inflatables have been proposed, ranging from general-purpose interfaces with dynamic shape changes to specific applications such as haptic sensation,

portable mobility, or soft robotics. In addition, shape-changing interfaces that construct structures using multiple inflatables [2, 9, 10] have been proposed as high-potential applications because they enable more complex shape changes by increasing the number of inflatable bladders.

However, there are still many challenges for inflatables in practical applications. Conventional pneumatic systems usually use air compressors for the large-sized systems and commercial pneumatic pumps for the tabletop-sized systems as the drive source. Both require external equipment as a drive source to feed air through tubing and are not self-contained systems. Most of these drive sources are noisy. In addition, most previous studies use active valves for pneumatic control, which further increases the drive noise. Furthermore, the conventional multi-inflatable system requires separate tubing and active valves for each bladder to allow air to flow in and out of each inflatable. These extra setups make the entire system bulky and complex, limiting the layout of the placement.

Several liquid-to-gas phase change actuators [3, 6–8], which utilize the volume change that occurs during the vaporization process, have been proposed in recent years. These actuators are not bulky owing to the absence of tubing, and can be driven silently and flexibly. However, most existing liquid-to-gas phase change actuators are made from small flat pouches, and three-dimensional pouches such as cubes are rarely made due to heating efficiency challenges. Furthermore, it is difficult to control the heating. It is limited to discrete on/off states, making it difficult to inflate programmatically to the middle states. Additionally, the module can be used only a limited number of times. Reconfigurability also needs to be improved because the design cannot be changed once the liquid is filled in advance. Thus, there are still many challenges in order to utilize liquid-to-gas phase change actuators in practical inflatable applications.

In this paper, we demonstrate InflatableMod[4], novel inflatable modules based on the efficient control of liquid-to-gas phase change actuators. These are designed with a compact circuit that has a liquid transfer function and a heating function. InflatableMod consists of Phase-change Control Module and Pouch Module. The Phase-change Control Module generates liquid flow and feeds a low-boiling-point liquid into the Pouch Module. The temperature of the liquid is controlled by a Peltier device, which causes the Pouch Module to inflate by the volume change of the phase change. We also utilize electrohydrodynamic (EHD) pumps [1], based on the EHD phenomenon, for the pump function to generate liquid flow because of their ability to drive quite silently. In addition, it has a high degree of freedom of placement and can be embedded into objects in different shapes. The EHD phenomenon generates liquid flow when a high voltage is applied to a dielectric liquid. Using an EHD pump, the liquid flow is generated simply by placing an electrode in the liquid and applying a voltage, eliminating the need for mechanical moving parts. The EHD pump design is based on HydroMod [5]. Furthermore, by feeding only the required amount of low-boiling-point liquid into the Pouch Module and then efficiently controlling the heating, the previously-mentioned drawbacks of existing liquid-to-gas phase change actuators can be mitigated and resolved. This feature enables the creation of inflatable systems of various shapes and sizes by liquid-to-gas phase change, as well as

control in the middle range. In addition, as each module is wirelessly synchronized, it is possible to create a multi-inflatable system in which each inflatable structure is untethered and can be reconfigured for placement. This allows inflatables to be placed anywhere without having to consider the location of tubing and valves, eliminating the limitation of the module layout and system complexity that have been the challenges for multi-inflatable.

In this paper, we demonstrate the basic operation and some applications with InflatableMod. The core novelty of InflatableMod lies in the first exploration of the novel inflatable approach in tabletop-sized inflatables free from extra device requirements, tubing constraints, and noises.

2 INFLATABLEMOD

2.1 System Overview

InflatableMod was designed based on the following requirements for use as versatile pneumatic physical interfaces:

- It can be driven as a compact, untethered, self-contained module in a simple system configuration that operates without needing bulky, heavy, and noisy equipment.
- It can be designed to be reconfigurable with inflatables of various shapes and sizes.
- It can be used repeatedly and can control inflatables in intermediate states.
- Multiple modules can be controlled wirelessly at the same time.

As shown in Fig. 2, InflatableMod consists of Phase-change Control Module and Pouch Module. The Phase-change Control Module has a liquid transfer function to generate liquid flow and feed low-boiling-point liquid into the Pouch Module and a liquid heating function to drive the liquid-to-gas phase change efficiently. The Pouch Module is a bladder that can be fabricated and inflated into various shapes and sizes depending on the application.

The operation of the inflatable system with the InflatableMod is conducted in six steps, as shown in Fig. 3. First, (A) a low-boiling-point liquid is injected into a small tank inside the Phase-change Control Module. Thereafter, (B) the Pouch Module and the Phase-change Control Module are attached together. (C) The inflatable



Figure 2: InflatableMod supports Phase-change Control Module and Pouch Module. Pouch Module have a variety of shapes and sizes, depending on the applications.

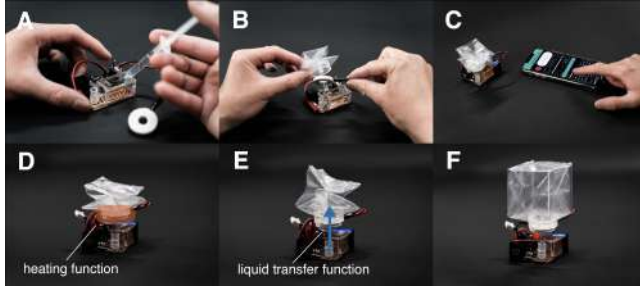


Figure 3: Six step process to operate an inflatable system with InflatableMod: (A) inject Novec 7000 into the tank of Phase-change Control Module, (B) connect the Pouch Module to the Phase-change Control Modules, (C) control the system with simple GUI software, (D) heat the reserving dish of Pouch Module, (E) feed low-boiling-point liquid into the Pouch Module, (F) inflate the Pouch Module.

system is then operated remotely using simple GUI software on a smartphone. (D) The liquid heating function of the Phase-change Control Module is then driven to heat the reservoir dish of the Pouch Module. Next, (E) the liquid transfer function is driven to feed the low-boiling-point liquid at an efficient timing to the reservoir dish of the Pouch Module that has just been heated. These operational flows allow the Pouch Module to be inflated by volume change on vaporization. InflatableMod uses Novec 7000 with a boiling point of 34°C as the low-boiling-point liquid.

Thus, the ability to drive inflatables with such a simple process allows them to be used in a highly versatile manner. We also point readers to the full paper [4] for efficient phase-change control methods.

2.2 Backflow prevention for programmable check valves with liquid transfer function

The liquid transfer control delivers liquid to the reservoir dish and significantly contributes as a programmable check valve. The InflatableMod delivers the required amount of low-boiling-point liquid to the Pouch Module, which is then heated to cause a liquid-to-gas phase change that inflates the pouch. However, because there is a flow path to feed the liquid into the pouch, simply heating the pouch causes the gas to flow back when it vaporizes. Therefore, to prevent backflow, the pump is not turned off during heating but is driven at a pressure below the threshold for pushing liquid into the reservoir dish. The output power of the check valve can be changed dynamically according to the pouch size and the amount of liquid to be injected. The EHD pump can prevent backflow without using a mechanical check valve.

2.3 Deflation Method

InflatableMod has two major deflation methods. The first method is deflation which uses a phase change from gas to liquid. This deflation method is shown in Fig. 4. The inflation flow is briefly described below (a-1~5). First, from the initial state (a-1), the low-boiling-point liquid is poured into Pouch Module (a-2,3). Then, the backflow prevention control described in 2.2 is activated. Because

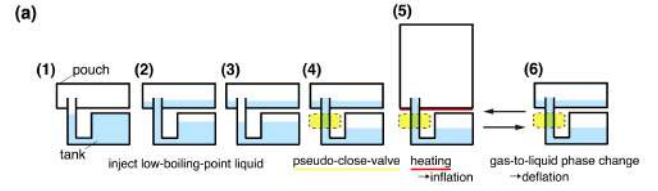


Figure 4: The pipeline of deflation method using a phase change from gas to liquid.

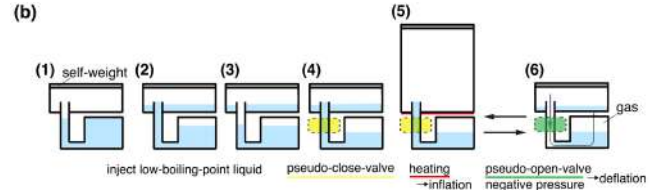


Figure 5: The pipeline of deflation method using self-weight and negative pressure.

the liquid pushed by the EHD Pump continuously stays in the channel connecting the tank and the pouch, the air cannot escape from the Pouch Module. This is considered as a pseudo-closed-valve state (a-4): the liquid in the channel functions as a stopper of the imaginary valve for sealing the gas inside. The low-boiling-point liquid is heated from the surface of the Pouch Module to induce a phase change and inflates the pouch (a-5). After the heating for inflation is stopped, the backflow prevention control is kept active for deflation operation (a-6). The inside of the pouch keeps the pseudo-close-valve state, and deflation occurs due to the volume change caused by the gas-to-liquid phase change.

The second method is deflation which uses self-weight and negative pressure. Fig. 5 shows the second deflation method. The inflation flow is described (b-1~5). This is the same as the first method (a-1~5). The first step is to attach a self-weight, such as butterfly wings shown in application 3.3, to the Pouch Module. Then, the liquid is returned to the Pump Unit with the backflow prevention turned off (b-6). Since the liquid no longer blocks the path that the air in the Pouch Module escapes, the inside of the pouch becomes a pseudo-open-valve state. The Pouch Module is first subjected to negative pressure and then to a release state in which no pressure is applied. Therefore, the self-weight causes the pouch to deflate in a short time. At this time, the gas inside the pouch escapes through the ventholes in the Phase-change Control Module's Pump Unit. This method deflates quickly and has a high affinity for representing kinetic motion. The actual deflation time is determined by the weight of the self-weight and the size of the Pouch Module. The heavier the self-weight or smaller the Pouch Module, the shorter the deflation time. In the biomimetic actuator shown in application 3.3, the self-weight of the butterfly wings is 8 g and the volume of the Pouch Module is approximate 25 mm x 40 mm x 10 mm. The deflation time in this case is approximately 10 seconds. Even if the pouch runs low on liquid after repeated deflation operations, it can be used repeatedly by simply filling it again from the tank in the Pump Unit of the Phase-change Control Module (b-2,3).

3 DEMO APPLICATIONS

We demonstrate the basic operation and some applications with InflatableMod.

3.1 Reconfigurable Inflatable Lamp

Fig. 6 shows a reconfigurable inflatable lamp that inflates and lights up and can be freely changed in shape according to user's mood. It is suitable for use in quiet places such as libraries or bedrooms. Furthermore, the inflatable lamp takes advantage of the long-time deflation feature of the phase change from gas to liquid. Application scenarios can be envisioned, such as studying until the inflatable lamp deflates, like a physical timer that adapts to the natural environment. It can be used in quiet work and living situations, such as a light for studying or for a nap.



Figure 6: Reconfigurable inflatable lamp. (A) Pouch modules come in a variety of shapes and sizes. (B) Various inflatable objects can be created dynamically by replacing the Pouch Module reconfigurably. (C) Reconfigurable inflatable lamp inflates and lights up.

3.2 Continuous Data Physicalization on a Map

Fig. 7 shows continuous inflatable data physicalization on a map. It is possible for the user to arrange multiple InflatableMod in any two-dimensional array and cover them with a map sheet. This allows the map to be visualized in 2.5 dimensions by inflating multiple InflatableMod to match the altitude of each terrain. It further shows the advantage of InflatableMod, which allows inflation in intermediate states. A pouch can be inflated to the desired volume by feeding low-boiling-point liquid into it. Users can decide where to place the InflatableMod and how much volume to inflate. As a result, it is easy to visualize other maps in 2.5 dimensions. In addition to being able to be used repeatedly, this application could potentially be used as an educational tool in the future.



Figure 7: Continuous data physicalization on a map. (A) Multiple InflatableMod are placed in any two-dimensional location. (B) Cover with a map sheet. (C) By inflating multiple InflatableMod for each terrain altitude, the map can be visualized in 2.5 dimensions.

3.3 Silent Biomimetic Soft Actuator

Fig. 8 shows a silent biomimetic soft actuator made using InflatableMod. It expresses the soft movement of a butterfly. Because the actuator is driven silently, it does not interfere with the natural

movement of the butterfly. When the wings of a butterfly rise, they move slowly in conjunction with inflation, and when the wings of a butterfly fall, they move quickly in conjunction with deflation using their self-weight and negative pressure. Here, the butterfly's wings are the self-weight. This inflection of the movement of the wings creates the reality of the butterfly's wing movement.

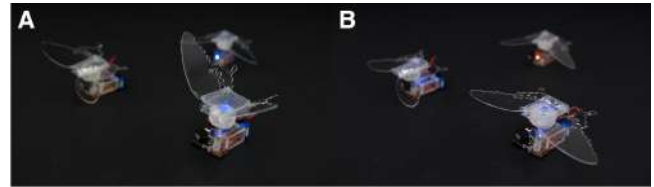


Figure 8: Silent biomimetic soft actuator. (A) The wings of a butterfly rise of inflation. (B) The wings of a butterfly fall of deflation using self-weight and passive pressure.

4 CONCLUSION

We demonstrate InflatableMod, novel inflatable modules based on the efficient control of liquid-to-gas phase change actuators. This modular design enables compact, silent, and untethered inflatable systems, solving the problems of existing tabletop-sized pneumatic systems that are bulky, heavy, and noisy. In addition, because the locations of tubing and valves no longer need to be considered, the layout restrictions that were considered difficult with multi-inflatable systems can be significantly alleviated. Furthermore, the drawbacks of conventional liquid-to-gas phase change actuators, such as controllability, operation speed, reusability, and reconfigurability, have been mitigated and resolved. InflatableMod could be the next step to breaking some new ground in inflatable technology. In the wake of this demonstration, we hope that various inflatables will become part of our daily lives.

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